International Data Encryption Algorithm (IDEA)

Module II

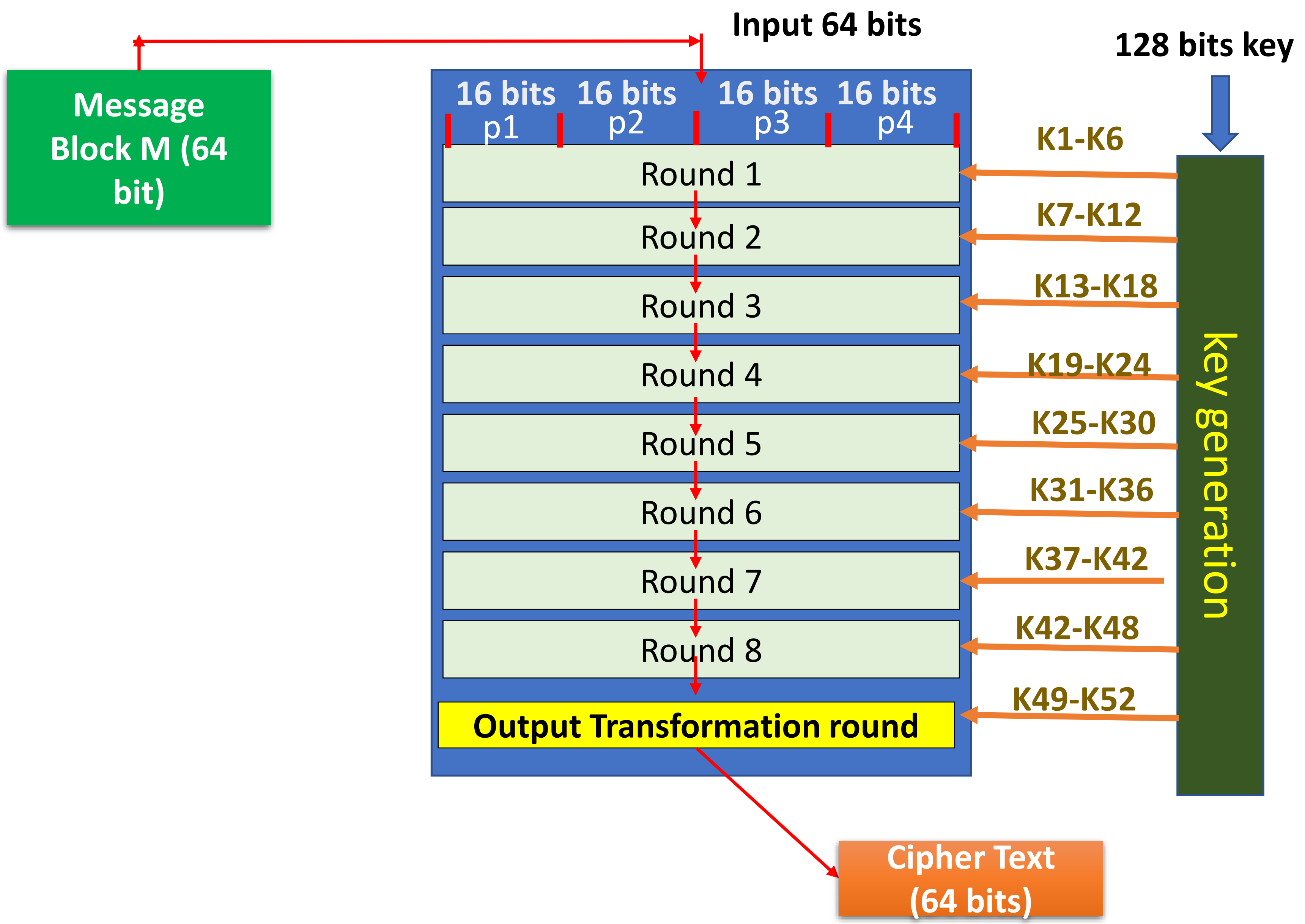
IDEA

- DES algorithm has been a popular secret key encryption algorithm and is used in many commercial and financial applications. However, its key size is too small by current standards and its entire 56 bit key space can be searched in approximately 22 hours
- IDEA is a block cipher designed by Xuejia Lai and James L. Massey in 1991
- It is a minor revision of an earlier cipher, PES (Proposed Encryption Standard)
- IDEA was originally called IPES (Improved PES) and was developed to replace DES.
- IDEA is similar to DES, both of them operates in rounds and both have complicated mangler function that does not need to be reversed for decryption.

IDEA

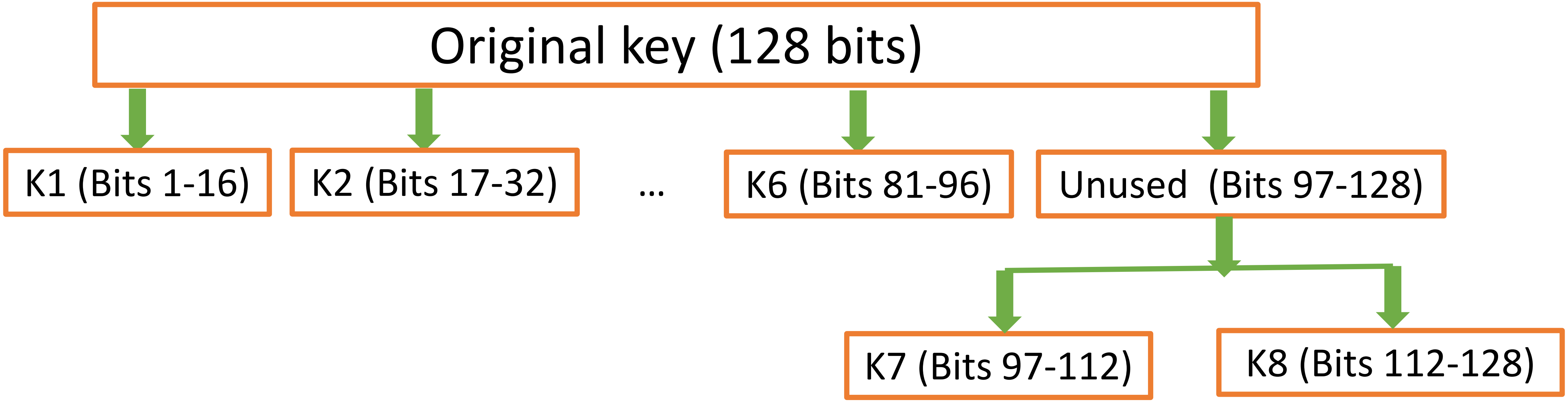
- Symmetric key block cipher.
- Key size is 128 bit from which 52 sub keys will be generated.
- Eight Identical transformation rounds.
- Each round 6 sub keys are used (16 bit each).
- Block size is 64 bits.
- In each round, block is divided into 4 portions/parts (16 bit each).
- One half round or output transformation round. It uses 4 sub keys (16 bit each). After this round gives ciphertext (64 bits).

Block diagram of IDEA



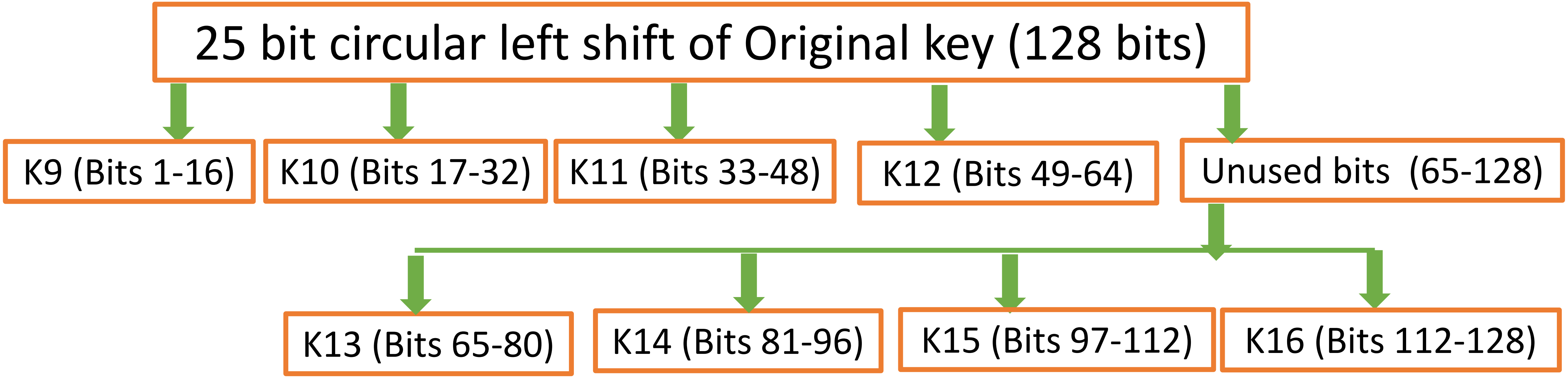
Key Generation in IDEA

Subkey generation for first round



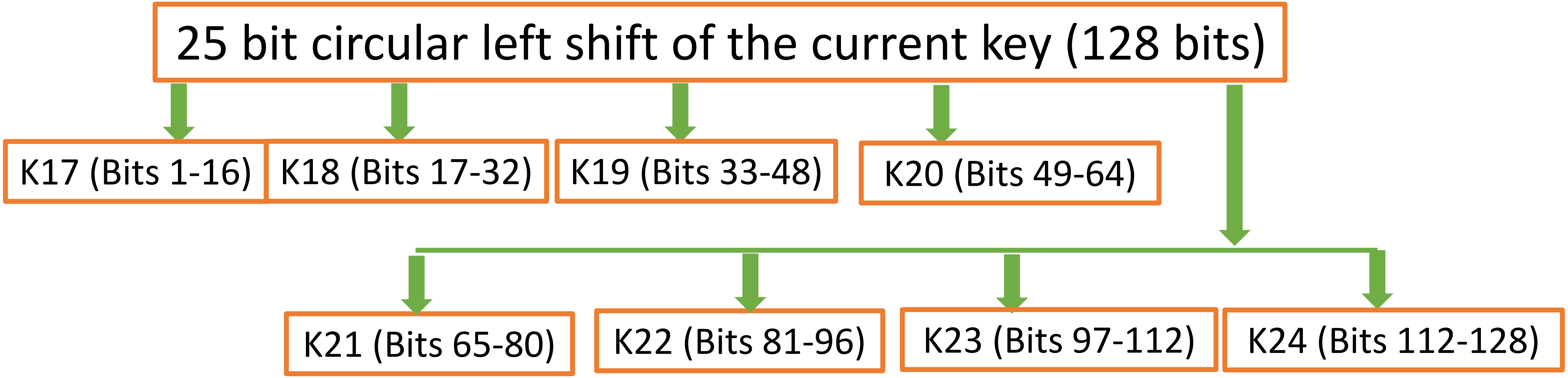
Key Generation in IDEA

Subkey generation for second round



Key Generation in IDEA

Subkey generation for third round



And it continues up to generation of k52.

Round function in IDEA

- $S1 = P1 \times K1$
- $S2 = P2 + K2$
- $S3 = P3 + K3$
- $S4 = P4 \times K4$
- $S5 = S1 \oplus S3$
- $S6 = S2 \oplus S4$
- $S7 = S5 \times K5$
- $S8 = S6 + S7$
- $S9 = S8 \times K6$
- $S10 = S7 + S9$

- $S11 = S1 \oplus S9$ => new P1 => p1
- $S12 = S3 \oplus S9$ => new P2 => p3
- $S13 = S2 \oplus S10$ => new P3 => p2
- $S14 = S4 \oplus S10$ => new P4 => p4

swap

The ooutput of each round given as input to the next round (p1 p3 p2 p4). Finally, at after round 8, the output is given to output Transformation without swapping (R1 R2 R3 R4).

Output transformation Phase

- R1 X K49 => C1
- R2 + K50 => C2
- R3 X K51 => C3
- R4 + K52 => C4

IDEA Decryption

The decryption process is exactly the same as the encryption process. There are some alterations in the generation and pattern of subkeys. The decryption subkeys are actually an Inverse of the encryption subkeys.